



**sdi**

Reference Manual

Version 1.18

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# 1 Overview

## Changelog:

### 1.18

- **Changed**
  - physicalports from 1.13 to 1.14
  - outputstatus from 1.14 to 1.15
  - portstatus from 1.14 to 1.15
  - jpegxsoutputstatus from 1.4 to 1.5

### 1.17

- **Changed**
  - physicalports from 1.12 to 1.13
  - outputstatus from 1.13 to 1.14
  - portstatus from 1.13 to 1.14
  - jpegxsoutputstatus from 1.3 to 1.4

### 1.16

- **Added**
  - inputRedundancy version 1.0
- **Changed**
  - physicalports from 1.11 to 1.12
  - outputstatus from 1.12 to 1.13
  - portstatus from 1.12 to 1.13
  - jpegxsoutputstatus from 1.2 to 1.3

### 1.15

- **Changed**
  - outputstatus from 1.11 to 1.12

### 1.14

- **Added**
  - component version 1.4
  - tsstatus version 1.3
- **Changed**
  - physicalports from 1.10 to 1.11
  - outputstatus from 1.10 to 1.11
  - portstatus from 1.11 to 1.12
  - jpegxsoutputstatus from 1.1 to 1.2
  - cardinfo from 1.1 to 1.2

### 1.13

- **Changed**

- physicalports from 1.9 to 1.10
- outputstatus from 1.9 to 1.10
- portstatus from 1.10 to 1.11
- jpegxsoutputstatus from 1.0 to 1.1
- transceiverStatus from 1.0 to 1.1

## 2 cardinfo (1.2)

### 2.1 Overview

Changelog:

#### 1.2

- **Added**
  - The field `hasTsEncapsulation` was added to `CardInfo` to support TS encapsulation.
- **Changed**
  - The field `isTicoXS` in `CardInfo` was renamed to `isJpegXS`

#### 1.1

- **Added**
  - The field `isTicoXS` was added to `CardInfo` to support Tico XS.

### 2.2 Command Reference

#### 2.2.1 GetCardInfo

- message `GetCardInfo.Request`
- message `GetCardInfo.Response`

### 2.3 Type Reference

#### 2.3.1 CardInfo

**struct**

|                                  |                     |
|----------------------------------|---------------------|
| <code>displayName</code>         | <code>string</code> |
| <code>isOptical</code>           | <code>bool</code>   |
| <code>hasOutputsOnly</code>      | <code>bool</code>   |
| <code>hasInputsOnly</code>       | <code>bool</code>   |
| <code>supportsPassthrough</code> | <code>bool</code>   |
| <code>isTicoHD</code>            | <code>bool</code>   |
| <code>isTicoUHD</code>           | <code>bool</code>   |
| <code>isJpegXS</code>            | <code>bool</code>   |
| <code>hasTsEncapsulation</code>  | <code>bool</code>   |

#### 2.3.2 CardInfoRequest

empty **struct**

#### 2.3.3 CardInfoResponse

**struct**

|                   |                       |
|-------------------|-----------------------|
| <code>data</code> | <code>CardInfo</code> |
|-------------------|-----------------------|



#### 2.3.4 GetCardInfo.Request

##### CardInfoRequest

#### 2.3.5 GetCardInfo.Response

##### CardInfoResponse

## 3 component (1.4)

### 3.1 Overview

Changelog:

#### 1.4

- Added
  - str2ComponentType mapping, the inverse of componentType2str.
- Changed
  - Added DVBTMLSUB and JPEG\_XS\_VIDEO to ComponentType type.

### 3.2 Type Reference

#### 3.2.1 AnyComponent

empty **struct**

#### 3.2.2 ComponentGroup

ComponentGroup: Grouping of ComponentType, used for filtering

**enum**

UNDEFINED  
AUDIO  
VIDEO  
SUBTITLE  
DATA

#### 3.2.3 ComponentLanguage

**string**

#### 3.2.4 ComponentSelector

Component selector.

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| @param pid      | Selects a component using a specific (source) PID.        |
| @param language | Selects a component by a three character language string. |
| @param type     | Selects a component by its type.                          |
| @param group    | Selects a component by group membership.                  |
| @param any      | Selects any component.                                    |

**variant**

|                |                          |
|----------------|--------------------------|
| pid            | <b>Pid</b>               |
| language       | <b>ComponentLanguage</b> |
| componentType  | <b>ComponentType</b>     |
| componentGroup | <b>ComponentGroup</b>    |
| any            | <b>AnyComponent</b>      |

### 3.2.5 ComponentType

A ComponentType is typically identified by a stream type (see ISO 13818-1 Table 2-29), often distinguished with another header field such as a descriptor tag.

#### enum

|                |  |
|----------------|--|
| MPEG_VIDEO     |  |
| MPEG_AUDIO     |  |
| TTX            |  |
| DVBSUB         |  |
| ECM            |  |
| PCR            |  |
| PRIVATE        |  |
| AC_3           |  |
| H264           |  |
| VBI            |  |
| AAC            |  |
| VC_1           |  |
| SCS_EMM        |  |
| DPI_CUE        |  |
| EAC3           |  |
| AIT            |  |
| HBBTV_CAROUSEL |  |
| DATA_CAROUSEL  |  |
| H265           |  |
| S302M          |  |
| JPEG2000       |  |
| S2038          |  |
| DVBTTMLSUB     |  |
| JPEG_XS_VIDEO  |  |

### 3.2.6 Pid

#### int

### 3.2.7 SubtitleFormat

#### enum

|                     |  |
|---------------------|--|
| DVB                 |  |
| DVB_HARD_OF_HEARING |  |
| EBU                 |  |
| EBU_HARD_OF_HEARING |  |

## 4 firewallTypes (1.0)

### 4.1 Type Reference

#### 4.1.1 IPAddress

string

#### 4.1.2 PaginatedRequest

Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request

@param pageNumber what page of the set to returned, determined by the slice [pageSize\*pageNumber

struct

|            |     |
|------------|-----|
| pageSize   | int |
| pageNumber | int |

#### 4.1.3 PaginationInfo

struct

|              |     |
|--------------|-----|
| totalEntries | int |
|--------------|-----|

the total number of entries on the server, before matching

#### 4.1.4 SocketAddress

Tuple with IP address and port

struct

|         |                                 |
|---------|---------------------------------|
| address | IPAddress                       |
|         | IP address, either IPv4 or IPv6 |
| port    | int                             |

#### 4.1.5 UUID

string

## 5 inputRedundancy (1.0)

### 5.1 Type Reference

#### 5.1.1 IdenticalSources

##### struct

|               |                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------|
| bufferDelay   | <b>int</b><br>Delay in milliseconds for offset handling between streams.                                     |
| guardInterval | <b>int</b><br>Delay in milliseconds used to compensate for fluctuations in the time offset between the flows |
| latencyTuning | <b>LatencyTuning</b><br>Selected latency tuning strategy                                                     |

#### 5.1.2 LatencyTuning

Enum describing desired latency tuning strategy

##### enum

|             |                                                        |
|-------------|--------------------------------------------------------|
| Auto        | The first flow activated will be assumed to be leading |
| MainLeading | The Main source is assumed to be leading               |
| MainLagging | The Main source is assumed to be lagging               |

#### 5.1.3 SliplessSwitching

##### struct

|                  |                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| identicalSources | <b>optional IdenticalSources</b><br>When the sources are expected to be identical, we allow more settings to improve latency performance                                      |
| switchOnTei      | <b>bool</b><br>When enabled, FPGA module additionally uses TEI status to assess alternative health. Available only for slipless fpga switching. Only one alternative allowed. |

## 6 jpegxoutputstatus (1.5)

### 6.1 Overview

#### Changelog:

##### 1.5

- **Changed**
  - bump physicalports version from 1.13 to 1.14.

##### 1.4

- **Changed**
  - bump physicalports version from 1.12 to 1.13.

##### 1.3

- **Changed**
  - bump physicalports version from 1.11 to 1.12.

##### 1.2

- **Added**
  - import physicalports version 1.11 has been added.
  - import tsstatus version 1.3 has been added.
  - JpegXSNetworkDelayStatus structure has been added.
  - XsTsOutputStatus structure has been added.
  - Added rpc GetJpegXSNetworkDelayStatus.
  - Added rpc GetTsStatus.
  - Added rpc GetXsTsOutputStatus.
  - Added Protocol GetJpegXSNetworkDelayStatusProtocol.
  - Added Protocol GetTsStatusProtocol.
  - Added Protocol XsTsOutputStatus.

##### 1.1

- **Added**
  - The structure JpegXSOffsetStatus was added.
  - Added RPC GetJpegXSOffsetStatus.
  - Added Protocol OffsetStatus.
- **Changed**
  - The field JpegXsAudioStatus in JpegXSAudioStatus function description was renamed to JpegXSAudioStatus.

### 6.2 Command Reference

### 6.2.1 GetJpegXSAudioStatus

RPCs

- message `GetJpegXSAudioStatus.Request`
- message `GetJpegXSAudioStatus.Response`

### 6.2.2 GetJpegXSOffsetStatus

- message `GetJpegXSOffsetStatus.Request`
- message `GetJpegXSOffsetStatus.Response`

### 6.2.3 GetJpegXSNetworkDelayStatus

- message `GetJpegXSNetworkDelayStatus.Request`
- message `GetJpegXSNetworkDelayStatus.Response`

### 6.2.4 GetTsStatus

- message `GetTsStatus.Request`
- message `GetTsStatus.Response`

### 6.2.5 GetXsTsOutputStatus

- message `GetXsTsOutputStatus.Request`
- message `GetXsTsOutputStatus.Response`

## 6.3 Type Reference

### 6.3.1 GetJpegXSAudioStatus.Request

empty `struct`

### 6.3.2 GetJpegXSAudioStatus.Response

`struct`

data `map` from `int` to `JpegXSAudioStatus`

### 6.3.3 GetJpegXSNetworkDelayStatus.Request

empty `struct`

### 6.3.4 GetJpegXSNetworkDelayStatus.Response

`struct`

data `map` from `int` to `JpegXSNetworkDelayStatus`

### 6.3.5 GetJpegXOffsetStatus.Request

empty **struct**

### 6.3.6 GetJpegXOffsetStatus.Response

**struct**

data **map** from **int** to **JpegXOffsetStatus**

### 6.3.7 GetTsStatus.Request

empty **struct**

### 6.3.8 GetTsStatus.Response

**struct**

data **map** from **UUID** to **tsstatus.TransportStreamStatus**

### 6.3.9 GetXsTsOutputStatus.Request

empty **struct**

### 6.3.10 GetXsTsOutputStatus.Response

**struct**

data **map** from **int** to **XsTsOutputStatus**

### 6.3.11 JpegXSAudioStatus

JpegXSAudioStatus

**struct**

|          |                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------|
| CH1Sync  | <b>optional bool</b><br>Optional bool, if not nullopt the audio channel is active, if true, it is also synced. |
| CH2Sync  | <b>optional bool</b>                                                                                           |
| CH3Sync  | <b>optional bool</b>                                                                                           |
| CH4Sync  | <b>optional bool</b>                                                                                           |
| CH5Sync  | <b>optional bool</b>                                                                                           |
| CH6Sync  | <b>optional bool</b>                                                                                           |
| CH7Sync  | <b>optional bool</b>                                                                                           |
| CH8Sync  | <b>optional bool</b>                                                                                           |
| CH9Sync  | <b>optional bool</b>                                                                                           |
| CH10Sync | <b>optional bool</b>                                                                                           |
| CH11Sync | <b>optional bool</b>                                                                                           |
| CH12Sync | <b>optional bool</b>                                                                                           |
| CH13Sync | <b>optional bool</b>                                                                                           |



|          |               |
|----------|---------------|
| CH14Sync | optional bool |
| CH15Sync | optional bool |
| CH16Sync | optional bool |

### 6.3.12 JpegXSNetworkDelayStatus

JpegXSNetworkDelayStatus

#### struct

|           |                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------|
| ptpActive | <b>bool</b><br>Is true if ptp is active on the box.                                               |
| video     | <b>int</b><br>The video essence network delay.                                                    |
| anc       | <b>int</b><br>The Ancillary essence newtwork delay.                                               |
| audios    | <b>map</b> from <b>int</b> to <b>int</b><br>A map of port number to audio essence network delays. |

### 6.3.13 JpegXSOffsetStatus

JpegXSOffsetStatus

#### struct

|        |                                                                                           |
|--------|-------------------------------------------------------------------------------------------|
| video  | <b>int</b><br>The video essence delay.                                                    |
| anc    | <b>int</b><br>The Ancillary essence delay.                                                |
| audios | <b>map</b> from <b>int</b> to <b>int</b><br>A map of port number to audio essence delays. |

### 6.3.14 XsTsOutputStatus

#### struct

|                 |                             |
|-----------------|-----------------------------|
| videoMode       | <b>physicalports.Format</b> |
| videoFractional | <b>bool</b>                 |
| videoUHD        | <b>bool</b>                 |
| imageWidth      | <b>int</b>                  |
| imageHeight     | <b>int</b>                  |
| frameRate       | <b>int</b>                  |
| syncLossCount   | <b>int</b>                  |

## 7 outputstatus (1.15)

### 7.1 Overview

#### Changelog:

##### 1.15

- **Changed**
  - bump physicalports version from 1.13 to 1.14.

##### 1.14

- **Changed**
  - bump physicalports version from 1.12 to 1.13.

##### 1.13

- **Added**
  - The fields `effectiveIdenticalSourceOffset`, `mainBufferDelay`, `backupBufferDelay` were added to `OutputSourceStatus` to support Hot Input Redundancy.
- **Changed**
  - bump physicalports version from 1.11 to 1.12.

##### 1.12

- **Added**
  - Added `AlternativeStatus` structure.
  - Added `OutputSourceStatus` structure.
  - The field `outputSources` was added to `SdiOutputStatus`.

##### 1.11

- **Changed**
  - bump physicalports version from 1.10 to 1.11.

##### 1.10

- **Changed**
  - bump physicalports version from 1.9 to 1.10.

##### 1.9

- **Changed**
  - bump physicalports version from 1.8 to 1.9.

##### 1.8

- **Changed**
  - bump physicalports version from 1.7 to 1.8.

##### 1.7

- **Changed**
  - bump physicalports version from 1.6 to 1.7.

##### 1.6

- **Changed**
  - bump physicalports version from 1.5 to 1.6.

## 1.5

- **Changed**
  - bump physicalports version from 1.4 to 1.5.

## 1.4

- **Changed**
  - bump physicalports version from 1.3 to 1.4.

## 1.3

- **Changed**
  - bump physicalports version from 1.2 to 1.3.

## 1.2

- **Changed**
  - bump physicalports version from 1.1 to 1.2.

## 1.1

- **Changed**
  - bump physicalports version from 1.0 to 1.1.

## 7.2 Command Reference

### 7.2.1 GetSdiOutputStatus

RPCs

- message **GetSdiOutputStatus.Request**
- message **GetSdiOutputStatus.Response**

## 7.3 Type Reference

### 7.3.1 AlternativeStatus

**struct**

isBroken **bool**

### 7.3.2 ClockFrequency

**enum**

NO\_TIME\_STAMP

CF\_27\_MHZ

CF\_1485\_MHZ

CF\_1485\_1001\_MHZ

CF\_297\_MHZ

CF\_297\_1001\_MHZ

### 7.3.3 Fec

#### enum

|             |  |
|-------------|--|
| NO_FEC      |  |
| L_FEC       |  |
| L_ADN_D_FEC |  |

### 7.3.4 Frame

#### enum

|                 |  |
|-----------------|--|
| FRAME_480I      |  |
| FRAME_576I      |  |
| FRAME_1080I     |  |
| FRAME_1080P     |  |
| FRAME_1080I_PSF |  |
| FRAME_2KP       |  |
| FRAME_2KI_PSF   |  |
| FRAME_720P      |  |
| UNKNOWN         |  |

### 7.3.5 Frate

#### enum

|                  |  |
|------------------|--|
| FR_2970_GHZ      |  |
| FR_2970_1001_GHZ |  |
| FR_1485_GHZ      |  |
| FR_1485_1001_GHZ |  |
| FR_0270_GHZ      |  |
| FR_60            |  |
| FR_60_1001       |  |
| FR_50            |  |
| FR_48            |  |
| FR_48_1001       |  |
| FR_30            |  |
| FR_30_1001       |  |
| FR_25            |  |
| FR_24            |  |
| FR_24_1001       |  |
| UNKNOWN          |  |

### 7.3.6 GetSdiOutputStatus.Request

empty **struct**

### 7.3.7 GetSdiOutputStatus.Response

#### struct

data

map from UUID to SdiOutputStatus

### 7.3.8 Map

#### enum

DIRECT\_SAMPLE

SMPTE\_B\_DL

SMPTE\_B\_DS

### 7.3.9 OutputSourceStatus

#### struct

effectiveIdenticalSourceOffset optional int

mainBufferDelay optional int

backupBufferDelay optional int

currentAlternativeIdx int

alternativeStatuses list of AlternativeStatus

### 7.3.10 ReferenceTime

#### enum

NOT\_LOCKED

RESERVED

LOCKED\_UTC

LOCKED\_PRIVATE

### 7.3.11 Sample

#### enum

UNKNOWN

S\_422\_10

S\_444\_10

S\_4444\_10

RESERVED

S\_422\_12

S\_444\_12

S\_4444\_12

S\_4224\_12

### 7.3.12 SdiOutputStatus

#### struct

videoMode physicalports.Format

missingPktCount int

|                 |                            |
|-----------------|----------------------------|
| syncLossCount   | int                        |
| extension       | int                        |
| formatPresent   | bool                       |
| vsid            | Vsid                       |
| frameCount      | int                        |
| refTime         | ReferenceTime              |
| scrambled       | bool                       |
| fec             | Fec                        |
| cf              | ClockFrequency             |
| streamStructure | Map                        |
| frame           | Frame                      |
| frate           | Frate                      |
| sample          | Sample                     |
| outputSources   | list of OutputSourceStatus |

### 7.3.13 Vsid

#### enum

|         |  |
|---------|--|
| PRIMARY |  |
| PROTECT |  |

## 8 physicalports (1.14)

### 8.1 Overview

#### Changelog:

##### 1.14

- **Added**
  - Added CHECKFIELD to Pattern enum that can be used to generate a pathological pattern.

##### 1.13

- **Added**
  - The enum PortCapacity was added to specifies the maximum format capacity of the port.
  - The field portCapacity was added to the PhysicalPort structure.
  - The rpc DeletePhysicalPorts was added to support deleting settings for one or more physical ports.

##### 1.12

- **Added**
  - The import inputRedundancy version 1.0 was added.
  - The field sliplessSwitching were added to the InputRedundancy structure.

##### 1.11

- **Added**
- The structure TsAudioSettings was added to support audio configuration for inputs on a JPEG XS TS product.
- The structure TsVideoSettings was added to support video configuration for inputs on a JPEG XS TS product.
- The structure TsAncSettings was added to support ancillary configuration for inputs on a JPEG XS TS product.
- The structure TsServiceSettings was added to support service configuration for inputs on a JPEG XS TS product.
- The structure TsSettings was added to support source configuration for inputs on a JPEG XS TS product.
- The enum FlowType was added to support Flow types.
- The fields flowType and tsSettings were added to InputSettings structure.
- The structure TsDecAudio was added to support information used to define each audio on the jpeg xs ts decoder.
- The structure TsDecAudioSettings was added to define all audios on a jpeg xs ts decoder card.
- The fields flowType and tsDecAudioSettings were added to OutputSettings structure.

##### 1.10

- **Added**
- The enum FrameSync was added to specify which reference signal we want to synchronize our video output to.
- The field frameSync was added to PhysicalPort structure .

- The fields `chStart` and `chWidth` were added to `ST2110Audio` structure.
- **Removed**
  - The field `channel` was removed from `ST2110Audio` structure .

## 1.9

- **Added**
  - The value `CUSTOM` was added to enum `TicoVideoCompressionRatio` .
  - The field `bitrate` was added to `TicoVideoEncodeConfiguration` structure.
  - The rpc `ClearAllCounters` was added.
  - The protocol `ClearAllCountersProtocol` was added.

## 1.8

- **Added**
  - The value `TEST_GENERATOR` was added to enum `Pattern` .
  - The enum `InputRedundancySwitchMode` was added to specify the `switchMode` between alternative sources.
  - The structure `PreferredSource` was added to specify preferred source configuration.
  - The structure `InputRedundancy` was added to specify data structure related to Input Redundancy feature.
  - The field `inputRedundancy` was added to `OutputSettings` structure.
- **Changed**
  - The enum `Format` was updated with ASI value and names of the VM1080lxx values.
  - The enum `AudioChannelRangeSelection` was updated with new values.

## 1.7

- **Added**
  - The enum `TextPosition` was added to specify the vertical placement for the text.
  - The structure `Color` was added to support color defined in terms of RGB.
  - The enum `BitmapBackgroundProperty` was added to specifies the color property of the background in a bitmap.
  - The structure `TextOverlay` was added to support text overlay to put on top of a test image.
  - The structure `MovingBox` was added to support moving box to be used in a test image generator.
  - The structure `TestGenerator` was added to defines how to generate a test image for a specific port.
  - The enum `TrafficPolicing` was added to specifies the different modes that are available for the trafficPolicing.
  - The field `testGenerator` and `trafficPolicing` were added to the `PhysicalPort` structure.
- **Changed**
  - The enum was renamed from `ExpectedFormat` to `Format` and updated with formats VM480P and VM576P.
  - The enum was renamed from `NoSignalPattern` to `Pattern`.

## 1.6



- **Added**

- The field `bufferDelay` from `ST2110GroupSource` structure was added.
- The structure `ST2110Audio` was added.

- **Changed**

- The field `anc` from `ST2110GroupSource` structure was changed from `uuid` to `optional(uuid)` type.
- The field `audio` from `ST2110GroupSource` structure was changed from `list(optional(uuid))` to `list(ST2110Audio)` type.

## 1.5

- **Added**

- The enum `NoSignalPattern` was added.
- The structure `ST2110GroupSource` was added.
- The field `noSignalPattern` was added to the `PhysicalPort` structure.
- The field `groupSource` was added to the `OutputSettings` structure.

## 1.4

- **Changed**

- The enumeration `PortDirection` has been updated with `Quad` values.

- **Removed**

- The protocol `SetPhysicalPortProtocol` and `DeletePhysicalPortProtocol` were removed.

## 1.3

- **Added**

- The protocol `SetPhysicalPortProtocol` and `DeletePhysicalPortProtocol` were added.

- **Changed**

- The enumeration `AudioChannelRangeSelection` has been updated.

## 1.2

- **Added**

- The enumeration `AudioInterval` was added to support time between audio packets.
- The enumeration `AudioEncapsulation` was added to support encapsulation format of the encoded audio channel.
- The enumeration `AudioChannelRangeSelection` was added to specifies input channels.
- The structure `AudioMapping` was added to support encoded audio channel.
- The enumeration `TicoVideoCompressionRatio` was added to specifies compression strength of the encoded video given in ratios.
- The enumeration `TicoVideoOptimization` was added to specifies what the encoded video should be optimized for.
- The structure `TicoVideoEncodeConfiguration` was added to specifies options for the TICO encoder.
- The fields `ticoConfiguration` and `audioMappings` were added to the `InputSettings`.

- **Changed**

- The type of the input field in PortSettings variant was change from optional(InputSettings) to InputSettings
- The type of the output field in PortSettings variant was change from optional(OutputSettings) to OutputSettings

## 1.1

- **Added**
  - The field id was added to the PhysicalPort.

## 8.2 Command Reference

### 8.2.1 GetPhysicalPorts

RPC for retrieving information regarding all ports, or only specific one if a filter is specified.

- message **GetPhysicalPorts.Request**
- message **GetPhysicalPorts.Response**

### 8.2.2 SetPhysicalPorts

RPC for submitting settings for one or more physical ports.

- message **SetPhysicalPorts.Request**
- message **SetPhysicalPorts.Response**

### 8.2.3 DeletePhysicalPorts

RPC for deleting settings for one or more physical ports. Should never be used.

- message **DeletePhysicalPorts.Request**
- message **DeletePhysicalPorts.Response**

### 8.2.4 ClearAllCounters

- message **ClearAllCounters.Request**
- message **ClearAllCounters.Response**

## 8.3 Type Reference

### 8.3.1 AudioChannelRangeSelection

Specifies which input channels should be used.

#### enum

|        |                         |
|--------|-------------------------|
| e1_2   | Use channels 1 and 2.   |
| e3_4   | Use channels 3 and 4.   |
| e5_6   | Use channels 5 and 6.   |
| e7_8   | Use channels 7 and 8.   |
| e9_10  | Use channels 9 and 10.  |
| e11_12 | Use channels 11 and 12. |
| e13_14 | Use channels 13 and 14. |
| e15_16 | Use channels 15 and 16. |

|        |                                         |
|--------|-----------------------------------------|
| e1_4   |                                         |
| e5_8   |                                         |
| e9_12  |                                         |
| e13_16 |                                         |
| e1_6   | Use channels 1, 2, 3, 4, 5 and 6.       |
| e2_7   | Use channels 2, 3, 4, 5, 6 and 7.       |
| e3_8   | Use channels 3, 4, 5, 6, 7 and 8.       |
| e4_9   | Use channels 4, 5, 6, 7, 8 and 9.       |
| e5_10  | Use channels 5, 6, 7, 8, 9 and 10.      |
| e6_11  | Use channels 6, 7, 8, 9, 10 and 11.     |
| e7_12  | Use channels 7, 8, 9, 10, 11 and 12.    |
| e8_13  | Use channels 8, 9, 10, 11, 12 and 13.   |
| e9_14  | Use channels 9, 10, 11, 12, 13 and 14.  |
| e10_15 | Use channels 10, 11, 12, 13, 14 and 15. |
| e11_16 | Use channels 11, 12, 13, 14, 15 and 16. |
| e1_8   |                                         |
| e9_16  |                                         |

### 8.3.2 AudioEncapsulation

Encapsulation format of the encoded audio channel.

#### enum

|               |                               |
|---------------|-------------------------------|
| SMPTE_2110_30 | AES67 audio transport format. |
| SMPTE_2110_31 | AES3 audio transport format.  |

### 8.3.3 AudioInterval

Time between audio packets.

#### enum

|         |                 |
|---------|-----------------|
| e1_MS   | 1 millisecond   |
| e125_US | 125 nanoseconds |

### 8.3.4 AudioMapping

Definition of an encoded audio channel created from a selection of incoming audio channels.

#### struct

|                  |                                                                             |
|------------------|-----------------------------------------------------------------------------|
| interval         | <b>AudioInterval</b><br>The rate between audio packets.                     |
| encapsulation    | <b>AudioEncapsulation</b><br>The encapsulation format to utilize.           |
| channelSelection | <b>AudioChannelRangeSelection</b><br>The incoming audio channels to encode. |

### 8.3.5 BitmapBackgroundProperty

Specifies the color property of the background in a bitmap.

#### enum

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| BLACK       | Make the bitmap background solid black.                                                     |
| TRANSPARENT | Make the bitmap background transparent, such that the bitmap in essence becomes an overlay. |

### 8.3.6 ClearAllCounters.Request

empty **struct**

### 8.3.7 ClearAllCounters.Response

empty **struct**

### 8.3.8 Color

A color defined in terms of RGB.

#### struct

|       |                                                |
|-------|------------------------------------------------|
| red   | <b>int</b><br>Amount of red. Value 0 to 255.   |
| green | <b>int</b><br>Amount of green. Value 0 to 255. |
| blue  | <b>int</b><br>Amount of blue. Value 0 to 255.  |

### 8.3.9 DeletePhysicalPorts.Request

Message to send to card to delete one or more physical ports.

#### struct

|     |                                                                              |
|-----|------------------------------------------------------------------------------|
| ids | <b>list of UUID</b><br>A list of physical port UUIDs that should be deleted. |
|-----|------------------------------------------------------------------------------|

### 8.3.10 DeletePhysicalPorts.Response

A simple acknowledgement message to indicate success. If the request is faulty then an error is returned instead.

empty **struct**

### 8.3.11 FlowType

#### enum

|                         |
|-------------------------|
| UNCOMPRESSED_RTP_2022_6 |
| TICO_UHD_2022_6         |

JPEG\_XS\_2110

JPEG\_XS\_TS

### 8.3.12 Format

Specifies the expected format of a SDI video signal on either the input or the output.

#### enum

|                |                                                                                   |
|----------------|-----------------------------------------------------------------------------------|
| OFF            | This port is not expected to be used.                                             |
| VM480I         | (Video Mode) 480 horizontal lines, interlaced.                                    |
| VM576I         | 576 horizontal lines, interlaced.                                                 |
| VM720P24       | 720 horizontal lines, progressive scan, 24 frames per second.                     |
| VM720P24_1001  | 720 horizontal lines, progressive scan, 24/1.001=23.97 frames per second (NTSC).  |
| VM720P25       | 720 horizontal lines, progressive scan, 25 frames per second.                     |
| VM720P30       | 720 horizontal lines, progressive scan, 30 frames per second.                     |
| VM720P30_1001  | 720 horizontal lines, progressive scan, 30/1.001=29.97 frames per second (NTSC).  |
| VM720P50       | 720 horizontal lines, progressive scan, 50 frames per second.                     |
| VM720P60       | 720 horizontal lines, progressive scan, 60 frames per second.                     |
| VM720P60_1001  | 720 horizontal lines, progressive scan, 60/1.001=59.94 frames per second (NTSC).  |
| VM1080P24      | 1080 horizontal lines, progressive scan, 24 frames per second.                    |
| VM1080P24_1001 | 1080 horizontal lines, progressive scan, 24/1.001=23.97 frames per second (NTSC). |
| VM1080P25      | 1080 horizontal lines, progressive scan, 25 frames per second.                    |
| VM1080P30      | 1080 horizontal lines, progressive scan, 30 frames per second.                    |
| VM1080P30_1001 | 1080 horizontal lines, progressive scan, 30/1.001=29.97 frames per second (NTSC). |
| VM1080I25      | 1080 horizontal lines, interlaced, 25 frames per second.                          |
| VM1080I30      | 1080 horizontal lines, interlaced, 30 frames per second.                          |
| VM1080I30_1001 | 1080 horizontal lines, interlaced, 30/1.001=29.97 frames per second (NTSC).       |
| VM1080P50      | 1080 horizontal lines, progressive scan, 50 frames per second.                    |
| VM1080P60      | 1080 horizontal lines, progressive scan, 60 frames per second.                    |
| VM1080P60_1001 | 1080 horizontal lines, progressive scan, 60/1.001=59.94 frames per second (NTSC). |
| VM2160P50      | 2160 horizontal lines, progressive scan, 50 frames per second.                    |
| VM2160P60      | 2160 horizontal lines, progressive scan, 60 frames per second.                    |
| VM2160P60_1001 | 2160 horizontal lines, progressive scan, 60/1.001=59.94 frames per second (NTSC). |
| ASI            | ASI signal over SDI.                                                              |
| ANY            | Any format.                                                                       |

### 8.3.13 FrameSync

Containing information used to synchronize frames.

#### struct

#### FrameSyncMode

|               |                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| frameSyncMode | mode used to synchronize frames.                                                                                               |
| genlockOffset | <b>optional int</b><br>optional field for setting pulse delay, in case genlock is used to sync frames on JPEG XS 2110 decoder. |

### 8.3.14 FrameSyncMode

Specifies which reference signal we want to synchronize our video output to

#### enum

|          |                       |
|----------|-----------------------|
| DISABLED | Frame sync disabled   |
| GENLOCK  | Frame sync to genlock |
| PTP      | Frame sync to PTP     |

### 8.3.15 GetPhysicalPorts.Request

Request either all physical ports with an empty request, or only specific physical ports matching the specified filter.

#### variant

|        |                                                                             |
|--------|-----------------------------------------------------------------------------|
| filter | <b>GetPortFilter</b><br>The filter to utilize when fetching physical ports. |
|--------|-----------------------------------------------------------------------------|

### 8.3.16 GetPhysicalPorts.Response

Contains the requested physical ports.

#### struct

|      |                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------|
| data | <b>map</b> from <b>UUID</b> to <b>PhysicalPort</b><br>A map of ports returned by the request.                        |
| info | <b>firewallTypes.PaginationInfo</b><br>Meta information about the data - i.e the total number of items on the server |

### 8.3.17 GetPortFilter

Specifies a search for specific ports.

#### struct

|            |                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------|
| pagination | <b>optional firewallTypes.PaginatedRequest</b><br>Settings for splitting the response data into pages |
| filter     | <b>optional PortFilter</b><br>Search term used to filter the data before pagination                   |

### 8.3.18 InputRedundancy

Data structure related to Input Redundancy feature

#### struct

|                   |                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| mode              | <b>InputRedundancySwitchMode</b><br>Selected switch mode                                                                             |
| backups           | <b>list of UUID</b><br>List of alternative sources                                                                                   |
| preferredSource   | <b>optional PreferredSource</b><br>When set allows to override default source preferences                                            |
| switchbackDelay   | <b>int</b><br>Delay in seconds to wait before switching back to a more preferred alternative or MAIN source.                         |
| sliplessSwitching | <b>inputRedundancy.SliplessSwitching</b><br>The FPGA will handle the redundancy switching without glitches, given identical sources. |

### 8.3.19 InputRedundancySwitchMode

Enum related to SwitchMode between alternative sources

#### enum

|           |                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| Disabled  | InputRedundancy is inactive; only MAIN source is used.                                                              |
| Reverting | Automatically selects first healthy alternative source including MAIN source; i.e. no alarms detected on the source |

### 8.3.20 InputSettings

Settings specific to an input port.

#### struct

|                   |                                                                                                                                                                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inputId           | <b>UUID</b><br>Unique identifier for the input.                                                                                                                                                                                                                                    |
| flowType          | <b>FlowType</b><br>Type of the input port configured, depending on the product.                                                                                                                                                                                                    |
| ticoConfiguration | <b>optional TicoVideoEncodeConfiguration</b><br>If not set no TICO encoding will take place. Otherwise this represents the options for the encoder.                                                                                                                                |
| audioMappings     | <b>list of optional AudioMapping</b><br>Specifies how the 16 incoming audio channels should be grouped and encoded. Each individual incoming audio channel can be used in at most 1 mapping. The list always has a size of 8 with unused audio mappings being set to None.         |
| tsSettings        | <b>optional TsSettings</b><br>Used only if flowType=JPEG_TS, will be an empty optional on others. Specifies the TS source information for the input. If this is used, audioMappings field above should be set to empty list since TS related audio is configured inside this type. |

### 8.3.21 MovingBox

Specify the properties of a moving box to be used in a test image generator.

#### struct

|         |                                                                             |
|---------|-----------------------------------------------------------------------------|
| enabled | <b>bool</b><br>Include this moving box in the test image to be generated.   |
| color   | <b>optional Color</b><br>The Color (R, G, B) of the box. Default is orange. |

### 8.3.22 OutputFormat

Specifies the format of the output SDI signal.

#### enum

|    |                                         |
|----|-----------------------------------------|
| SD | Standard Definition output.             |
| HD | High Definition output, 720p or higher. |

### 8.3.23 OutputSettings

Settings specific to an output port.

#### struct

|                    |                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------|
| outputId           | <b>UUID</b><br>The unique identifier for the output.                                                                |
| source             | <b>UUID</b><br>The unique identifier for the source that will be used as output.                                    |
| groupSource        | <b>optional ST2110GroupSource</b><br>If set, the source is a ST2110 Group Source, and is identified with this UUID. |
| nextSource         | <b>optional UUID</b><br>•                                                                                           |
| sources            | <b>map</b> from <b>UUID</b> to <b>Service</b><br>A list of all the services that make up the output.                |
| format             | <b>OutputFormat</b><br>Specifies the format of the output.                                                          |
| inputRedundancy    | <b>InputRedundancy</b>                                                                                              |
| tsDecAudioSettings | <b>optional TsDecAudioSettings</b><br>Specifies the audio settings of the jpeg xs ts decoder.                       |
| flowType           | <b>FlowType</b><br>Type of the input port configured, depending on the product.                                     |

### 8.3.24 Pattern

Specifies an image pattern.

#### enum

|               |                                                                           |
|---------------|---------------------------------------------------------------------------|
| NONE          | Do not generate any image. Not available for TestGenerator.               |
| BLACK_PICTURE | Generate an entirely black frame.                                         |
| COLOR_BAR     | Generate an image consisting of several distinctive, vertical color bars. |



|                |                                                                                             |
|----------------|---------------------------------------------------------------------------------------------|
| FREEZE_FRAME   | Freeze on the last valid frame seen on the relevant port. Not available for Test Generator. |
| TEST_GENERATOR | Enable Test Generator when no signal detected.                                              |
| CHECKFIELD     | Generate a pathological image.                                                              |

### 8.3.25 PhysicalPort

Specifies all relevant settings relating to a single physical SDI port.

#### struct

|                 |                                                                                                                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| enabled         | <b>bool</b><br>Enables the port for input/output. If disabled, then the port is unusable.                                                                                                                                                     |
| portNumber      | <b>int</b><br>The number of the port. If it is electrical, then it is the logical, physical ordering (1,2,3,4,...) if optical, then the first port has a value of 4, and the second port has a value of 1.                                    |
| label           | <b>string</b><br>The user-modifiable nickname for this port, in order to easily distinguish it from the rest.                                                                                                                                 |
| expectedFormat  | <b>Format</b><br>The expected format of the SDI video this input/output port is assuming to receive. If there is a large discrepancy between the expected and actual format, then the traffic policer will kick in and mute the input/output. |
| portSettings    | <b>PortSettings</b><br>Defines direction-specific (input/output) port settings.                                                                                                                                                               |
| direction       | <b>PortDirection</b><br>Specifies the port direction. A port can either be an input or an output, not both.                                                                                                                                   |
| noSignalPattern | <b>Pattern</b><br>Specifies the image to be generated when there is a signal loss.                                                                                                                                                            |
| testGenerator   | <b>TestGenerator</b><br>Specifies the test image generator, which can be useful for initial signal testing. The test generator image can also be used as a noSignalPattern image if specified.                                                |
| trafficPolicing | <b>TrafficPolicing</b><br>Specifies a maximum rate for the input signal. If the set rate is exceeded the FPGA mutes the stream.                                                                                                               |
| frameSync       | <b>FrameSync</b><br>Containing information used to synchronize frames                                                                                                                                                                         |
| portCapacity    | <b>optional PortCapacity</b><br>Specifies the maximum format capacity of the port. Only used for XS TS Enc optical, which is why it's optional.                                                                                               |

### 8.3.26 PortCapacity

Specifies the maximum format capacity of the port

#### enum

|     |                                                                 |
|-----|-----------------------------------------------------------------|
| NO  | The port will not handle anything. In essence it is turned off. |
| HD  | The port can handle up to HD formats/bitrates                   |
| UHD | The port can handle up to UHD formats/bitrates                  |

### 8.3.27 PortDirection

Specifies the direction of the port.

#### enum

|                             |                                                                                                                                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INPUT                       | The port is to receive an ingoing SDI signal.                                                                                                                                                                        |
| OUTPUT                      | The port is to produce an outgoing SDI signal.                                                                                                                                                                       |
| PASSTHROUGH                 | The port is to pass a signal directly to another port on the same card.                                                                                                                                              |
| CLONED                      | The port is one of two or more ports on the card that sends an identical video.                                                                                                                                      |
| QUAD_INPUT                  | For UHD coding. Groups four HD signals (a quadrant each) into one UHD video.                                                                                                                                         |
| QUAD_OUTPUT_SQUARE_DIVISION | For UHD coding. Takes a UHD signal and splits it into four quadrants (four HD signals), going clockwise from the upper-left quadrant.                                                                                |
| QUAD_OUTPUT_INTERLEAVED     | For UHD coding. Uses 2-sample interleave. Groups of two pixels are distributed across four links, Even rows (further divided into two-pixel groups) are sent across link 1 and 2, odd rows are sent on link 3 and 4. |

### 8.3.28 PortFilter

Specifies a filter to be used as part of GetPhysicalPort RPCs and similar.

#### struct

searchTerm **optional string**

### 8.3.29 PortSettings

A physical port can be either input or output type port.

#### variant

|        |                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------|
| input  | <b>InputSettings</b><br>If PortSettings equal an InputSettings type, then this indicates that it is an input port.    |
| output | <b>OutputSettings</b><br>If PortSettings equal an OutputSettings type, then this indicates that it is an output port. |

### 8.3.30 PreferredSource

Preferred source configuration

#### struct

**int**

|        |                                                                                                                                                                                                                                                                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| index  | Selects a source that should be marked as preferred. Index = 0 is MAIN source, index = 1 means first backup, etc. Preferred source will be used as the default source when it is healthy. If it fails the system will switch using regular source priorities (for example: first MAIN, then first backup, then second backup and so on) |
| locked | <b>bool</b><br>When enabled disables switching to another source when preferred source is faulty                                                                                                                                                                                                                                        |

### 8.3.31 ST2110Audio

Defines a set of audio packets that are used as part of a ST2110 stream.

#### struct

|         |                                                                                |
|---------|--------------------------------------------------------------------------------|
| essence | <b>UUID</b><br>Uniquely identifies the audio.                                  |
| chStart | <b>int</b><br>Identifies to which channel map first audio channel from essence |
| chWidth | <b>int</b><br>Identifies how many channels is contained in the essence         |

### 8.3.32 ST2110GroupSource

Describes a logical grouping of ST2110 sources.

#### struct

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| bufferDelay | <b>int</b><br>Maximum allowable delay in milliseconds.                                      |
| isTico      | <b>bool</b><br>If flag is set, then the video is TICO-encoded.                              |
| video       | <b>UUID</b><br>Unique identifier for the video source.                                      |
| anc         | <b>optional UUID</b><br>Unique identifier for ancillary data packets.                       |
| audio       | <b>list of ST2110Audio</b><br>Defines the audio packets that are part of this group source. |

### 8.3.33 Service

Service information

#### struct

|           |                                               |
|-----------|-----------------------------------------------|
| name      | <b>string</b><br>The name of the service.     |
| sid       | <b>int</b><br>The service ID.                 |
| thumbnail | <b>string</b><br>The corresponding thumbnail. |

### 8.3.34 SetPhysicalPorts.Request

Message to send to card to set one or more physical port settings.

#### struct

|      |                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------|
| data | <b>map</b> from <b>UUID</b> to <b>PhysicalPort</b><br>A map of physical port UUID and the corresponding Physical-Port object to set. |
|------|--------------------------------------------------------------------------------------------------------------------------------------|

### 8.3.35 SetPhysicalPorts.Response

A simple acknowledgement message to indicate success. If the request is faulty then an error is returned instead.

empty **struct**

### 8.3.36 TestGenerator

Defines how to generate a test image for a specific port.

#### struct

|             |                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| force       | <b>bool</b><br>Immediately override the input/output signal on the port and display the test image with desired settings. If false, the test generated image will only display if it is set as the noSignal-Pattern on the PhysicalPort, and there is a signal loss. |
| format      | <b>Format</b><br>The format of the test image. Can be up to fullHD resolutions on inputs. Outputs can have resolutions up to ultraHD.                                                                                                                                |
| pattern     | <b>Pattern</b><br>The TestGeneratorPattern to generate as base image.                                                                                                                                                                                                |
| textOverlay | <b>TextOverlay</b><br>If flag is set, then the port label will be displayed on the test generated image.                                                                                                                                                             |
| movingBox   | <b>MovingBox</b><br>If set, a box with the desired color will be re-rendered on the test-image at a set interval to confirm that the signal has not frozen.                                                                                                          |
| audio       | <b>bool</b><br>If enabled, a constant audio tone will be generated alongside the test-generated image in order to confirm a continuous, non-interrupted audio signal.                                                                                                |

### 8.3.37 TextOverlay

Specify the properties of a text overlay to put on top of a test image.

#### struct

|         |                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| enabled | <b>bool</b><br>Enable the text overlay. The label of the port will be used as the text. If the text is too long for the image, it will be shortened. |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------|
| position   | <b>TextPosition</b><br>Specify where to place the text on the vertical plane. It will always be centered horizontally. |
| background | <b>BitmapBackgroundProperty</b><br>Specify the properties of the background in the bitmap.                             |

### 8.3.38 TextPosition

Specify the vertical placement for the text.

#### enum

|        |                                      |
|--------|--------------------------------------|
| TOP    | Place it at the top of the image.    |
| MIDDLE | Place it in the middle of the image. |
| BOTTOM | Place it at the bottom of the image. |

### 8.3.39 TicoVideoCompressionRatio

Compression strength of the encoded video given in ratios.

#### enum

|        |                             |
|--------|-----------------------------|
| e1_2   | Compress to a ratio of 1:2. |
| e1_4   | Compress to a ratio of 1:4. |
| e1_5   | Compress to a ratio of 1:5. |
| CUSTOM |                             |

### 8.3.40 TicoVideoEncodeConfiguration

Specifies options for the TICO encoder.

#### struct

|                  |                                                                                           |
|------------------|-------------------------------------------------------------------------------------------|
| compressionRatio | <b>TicoVideoCompressionRatio</b><br>The compression strength to utilize.                  |
| optimization     | <b>TicoVideoOptimization</b><br>Specifies what the encoded video should be optimized for. |
| bitrate          | <b>optional float</b>                                                                     |

### 8.3.41 TicoVideoOptimization

Specifies what the encoded video should be optimized for.

#### enum

|        |                                          |
|--------|------------------------------------------|
| PSNR   | Optimize for Peak Signal-to-Noise Ratio. |
| VISUAL | Optimize for perceived visual quality.   |

### 8.3.42 TrafficPolicing

Specifies the different modes that are available for the trafficPolicing

**enum**

|         |                                        |
|---------|----------------------------------------|
| TP_ZERO | Nothing shall pass through the policer |
| TP_SD   | Everything up to SD will pass through  |
| TP_HD   | Everything up to HD will pass through  |
| TP_3G   | Policer will allow everything          |

**8.3.43 TsAncSettings**

Ancillary configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

**struct**

|        |                                              |
|--------|----------------------------------------------|
| ancPid | <b>int</b><br>Ancillary pid for the service. |
|--------|----------------------------------------------|

**8.3.44 TsAudioSettings**

Audio configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

**struct**

|                  |                                                                             |
|------------------|-----------------------------------------------------------------------------|
| pid              | <b>int</b><br>Pid number of this audio channel in resulting TS stream.      |
| channelSelection | <b>AudioChannelRangeSelection</b><br>The incoming audio channels to encode. |

**8.3.45 TsDecAudio**

Containing information used to define each audio on the jpeg xs ts decoder.

**struct**

|         |                                                                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------|
| pid     | <b>int</b><br>number of this audio channel in resulting TS stream. When autoSelectPids is true, this value will be ignored. |
| chStart | <b>int</b><br>Identifies which channel the audio mapping will start on.                                                     |
| chWidth | <b>int</b><br>Identifies how many channels will be contained in the audio.                                                  |

**8.3.46 TsDecAudioSettings**

Settings to define all audios on a jpeg xs ts decoder card.

**struct**

|                |                                                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| autoSelectPids | <b>bool</b><br>Defines which allocation method to apply to pids for the audios. True means the the lowest numbered pid will be the first audio selected, false means filling out the pids manually. |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

audioMappings

**list** of **TsDecAudio**

Defines a list of the audios.

### 8.3.47 TsServiceSettings

Service configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

#### struct

serviceName

**string**

Name of the service.

serviceId

**int**

ID number of the service.

pmtPid

**int**

PMT pid for the service.

pcrPid

**int**

PCR pid for the service.

pcrInterval

**float**

Interval in which PCR is signalled repeatedly (in milliseconds).

### 8.3.48 TsSettings

Source configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

#### struct

tsServiceSettings

**TsServiceSettings**

Service settings for the source.

tsVideoSettings

**TsVideoSettings**

Video settings for the source.

tsAncSettings

**TsAncSettings**

Ancillary settings for the source.

tsAudioSettings

**list** of **optional TsAudioSettings**

Audio settings for the source.

### 8.3.49 TsVideoSettings

Video configuration for inputs on a JPEG XS TS product. This one is used for TS streams and supported on JPEG XS TS cards.

#### struct

videoPid

**int**

Video pid for the service.

## 9 portstatus (1.15)

### 9.1 Overview

Changelog:

#### 1.15

- **Changed**
  - physicalports from 1.13 to 1.14

#### 1.14

- **Changed**
  - physicalports from 1.12 to 1.13

#### 1.13

- **Changed**
  - physicalports from 1.11 to 1.12

#### 1.12

- **Changed**
  - physicalports from 1.10 to 1.11

#### 1.11

- **Changed**
  - physicalports from 1.9 to 1.10
  - transceiverStatus from 1.0 to 1.1

#### 1.10

- **Changed**
  - physicalports from 1.8 to 1.9

### 9.2 Command Reference

#### 9.2.1 GetPortStatus

- message **GetPortStatus.Request**
- message **GetPortStatus.Response**

### 9.3 Type Reference

#### 9.3.1 AudioChannelStatus

Status of an incoming audio channel.

**struct**

|              |                          |
|--------------|--------------------------|
| format       | <b>AudioFormat</b>       |
| samplingRate | <b>AudioSamplingRate</b> |



### 9.3.2 AudioFormat

Audio format of an incoming audio channel.

#### enum

|                    |
|--------------------|
| PCM                |
| DOLBY_DIGITAL      |
| DOLBY_DIGITAL_PLUS |
| DOLBY_E            |

### 9.3.3 AudioSamplingRate

Sampling rate in kHz.

#### enum

|                    |
|--------------------|
| SYNC_48            |
| ASYNC_48           |
| SYNC_44_1          |
| ASYNC_44_1         |
| SYNC_32            |
| ASYNC_32           |
| SYNC_FREE_RUNNING  |
| ASYNC_FREE_RUNNING |

### 9.3.4 GetPortStatus.Request

#### variant

|     |              |
|-----|--------------|
| ids | list of UUID |
|-----|--------------|

### 9.3.5 GetPortStatus.Response

#### struct

|      |                             |
|------|-----------------------------|
| data | map from UUID to PortStatus |
|------|-----------------------------|

### 9.3.6 PortStatus

PortStatus

#### struct

|               |                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lock          | bool                                                                                                                                                                                  |
| inputFormat   | physicalports.Format                                                                                                                                                                  |
| syncLostCount | int                                                                                                                                                                                   |
| edhErrorCount | int                                                                                                                                                                                   |
| audioStatuses | list of optional AudioChannelStatus<br>A list of length 16 corresponding to the statuses of the 16 audio channels. If no audio is detected is on the channel the status will be None. |

transceiverStatus

optional transceiverStatus.TransceiverStatus

## 10 quadinputstatus (1.0)

### 10.1 Command Reference

#### 10.1.1 GetQuadInputStatus

- message **GetQuadInputStatus.Request**
- message **GetQuadInputStatus.Response**

### 10.2 Type Reference

#### 10.2.1 GetQuadInputStatus.Request

empty **struct**

#### 10.2.2 GetQuadInputStatus.Response

**struct**

data **map** from **int** to **QuadInputStatus**

#### 10.2.3 QuadInputMode

**enum**

UNKNOWN  
SQUARE\_DIVISION  
INTERLEAVED

#### 10.2.4 QuadInputStatus

**struct**

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| instance      | <b>int</b><br>Ports 1-4 instance 0, 5-8 instance 1 ...              |
| hasLock       | <b>bool</b>                                                         |
| lockLossCount | <b>int</b>                                                          |
| mode          | <b>QuadInputMode</b><br>Mode recived from fpga                      |
| ancMode       | <b>QuadInputMode</b><br>Mode recived in ANC frame for this instance |
| skew          | <b>list of float</b><br>Skew for channel in ns                      |

## 11 ticooutputstatus (1.1)

### 11.1 Command Reference

#### 11.1.1 GetTicoOutputStatus

- message **GetTicoOutputStatus.Request**
- message **GetTicoOutputStatus.Response**

### 11.2 Type Reference

#### 11.2.1 AudioInputInfo

##### enum

|                                |  |
|--------------------------------|--|
| NO_AUDIO_DATA                  |  |
| AI_16CH_LINK_1                 |  |
| AI_16CH_LINK_2                 |  |
| AI_16CH_LINK_3                 |  |
| AI_16CH_LINK_4                 |  |
| AI_32CH_LINK_1                 |  |
| AI_32CH_LINK_2                 |  |
| AI_32CH_LINK_3                 |  |
| AI_32CH_LINK_4                 |  |
| AI_16CH_LINK_1_PLUS_16CH_LINK_ |  |
| AI_16CH_LINK_1_PLUS_16CH_LINK_ |  |
| AI_16CH_LINK_1_PLUS_16CH_LINK_ |  |
| AI_16CH_LINK_2_PLUS_16CH_LINK_ |  |
| AI_16CH_LINK_2_PLUS_16CH_LINK_ |  |
| AI_16CH_LINK_3_PLUS_16CH_LINK_ |  |

#### 11.2.2 ColorPrimaries

##### enum

|                    |  |
|--------------------|--|
| UNSPECIFIED        |  |
| ITU_R_BT601_7_D65  |  |
| ITU_R_BT709_6_D65  |  |
| ITU_R_BT2020_2_D65 |  |
| SMPTE_RP_431_2_DCI |  |
| SMPTE_RP_431_2_D65 |  |
| SRGB_D65           |  |

#### 11.2.3 FrameRate

##### enum

|                  |  |
|------------------|--|
| NO_DEFINED_VALUE |  |
| RESERVED         |  |
| FR_23_976        |  |

|             |
|-------------|
| FR_24       |
| FR_48_1001  |
| FR_25       |
| FR_29_97    |
| FR_30       |
| FR_48       |
| FR_50       |
| FR_59_64    |
| FR_60       |
| FR_100      |
| FR_120_1001 |
| FR_120      |

#### 11.2.4 GetTicoOutputStatus.Request

empty **struct**

#### 11.2.5 GetTicoOutputStatus.Response

**struct**

data **map** from **UUID** to **TicoOutputStatus**

#### 11.2.6 InterlaceMode

**enum**

|                         |
|-------------------------|
| PROGRESSIVE             |
| INTERLACED_FIRST_FIELD  |
| INTERLACED_SECOND_FIELD |

#### 11.2.7 MatrixCoefficient

**enum**

|                |
|----------------|
| UNSPECIFIED    |
| ITU_R_BT709_6  |
| ITU_R_BT601    |
| ITU_R_BT2020_2 |

#### 11.2.8 SampleBitdepth

**enum**

|            |
|------------|
| SB_8_BITS  |
| SB_10_BITS |
| SB_12_BITS |

### 11.2.9 SamplingStructure

#### enum

SS\_422\_YCBCR

SS\_444\_YCBCR

SS\_444\_GBR

SS\_420

### 11.2.10 SourceType

#### enum

UNKNOWN

SDI

NATIVE

### 11.2.11 TicoOutputStatus

#### struct

|                   |                   |
|-------------------|-------------------|
| instanceNumber    | int               |
| ticoDetected      | bool              |
| bitstreamProfile  | int               |
| frameRate         | FrameRate         |
| interlaceMode     | InterlaceMode     |
| sampleBitdepth    | SampleBitdepth    |
| samplingStructure | SamplingStructure |
| horizontalSize    | int               |
| verticalSize      | int               |
| colorPrimaries    | ColorPrimaries    |
| transferCharacter | TransferCharacter |
| matrixCoefficient | MatrixCoefficient |
| sourceType        | SourceType        |
| videoInputInfo    | VideoInputInfo    |
| audioInputInfo    | AudioInputInfo    |
| vbiInputInfo      | VbiInputInfo      |
| maxFifoFilling    | int               |

### 11.2.12 TransferCharacter

#### enum

UNSPECIFIED

ITU\_R\_BT601\_BT709\_6\_BT2020\_2

SMPTE\_ST\_2084\_2014

SLOG\_3

ARIB\_STD\_B67

### 11.2.13 VbiInputInfo

#### enum

NO\_VBI\_DATA

VBI\_LINK\_1

VBI\_LINK\_2

VBI\_LINK\_3

VBI\_LINK\_4

### 11.2.14 VideoInputInfo

#### enum

UNKNOWN

VI\_QUAD\_SQD\_LEVEL\_A

VI\_QUAD\_SQD\_LEVEL\_B

VI\_QUAD\_2SI\_LEVEL\_A

VI\_QUAD\_2SI\_LEVEL\_B

VI\_12G\_SDI

VI\_6G\_SDI

VI\_DUAL\_3G\_SDI

VI\_QUAD\_HD\_SDI

## 12 transceiverStatus (1.1)

### 12.1 Type Reference

#### 12.1.1 CableLengthIndicator

Information about the cable length indicator measurement. The measurement is given in terms of the Belden 1694A cable.

##### struct

|                 |                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| length          | <b>int</b><br>An indication of the cable length carrying the SDI input signal [m].                                                                                        |
| lengthPrecision | <b>int</b><br>The precision of the cable length measurement [m]. The actual length of the cable is likely to be in the interval [length - precision, length + precision]. |
| maxLength       | <b>int</b><br>The maximum supported cable length [m].                                                                                                                     |
| threshold       | <b>int</b><br>The threshold below which measurements of cable length are very imprecise [m].                                                                              |

#### 12.1.2 DataRate

The various input data rates the transceiver can lock onto.

##### enum

|               |
|---------------|
| DR_11_88_GBPS |
| DR_5_94_GBPS  |
| DR_2_97_GBPS  |
| DR_1_485_GBPS |
| DR_270_MBPS   |
| NO_LOCK       |

#### 12.1.3 SignalProperties

##### struct

|                      |                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| dataRate             | <b>DataRate</b><br>An indication of the data rate of the SDI signal that the transceiver has locked.                                          |
| cableLength          | <b>optional CableLengthIndicator</b><br>An indication of the cable length carrying the SDI input signal [m]. N/A if dataRate is NO_LOCK.      |
| horizontalEyeOpening | <b>optional float</b><br>Horizontal eye opening of the equalized SDI signal [UI]. Not available if dataRate is NO_LOCK. Not available for SD. |



#### 12.1.4 TransceiverStatus

Status about the quality of an SDI signal. These measurements are taken by the transceiver which is connected to the physical connector carrying the SDI signal.

##### struct

|                  |                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------|
| carrierDetect    | <b>bool</b><br>True if the transceiver has detected that there is an input signal from an SDI cable.            |
| signalProperties | <b>optional SignalProperties</b><br>Properties of the input signal that are available if a carrier is detected. |

## 13 tsstatus (1.3)

### 13.1 Overview

Changelog:

#### 1.3

- Changed
  - Dependency module component bumped from version 1.1 to 1.4

### 13.2 Type Reference

#### 13.2.1 OutputTransportStreamStatus

Status for the output TS

**struct**

|          |                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| services | <b>map</b> from <b>int</b> to <b>ServiceSourceStatus</b><br>Status for each service in the TS. Map from service ID to status struct. |
| pids     | <b>list</b> of <b>PidStatus</b><br>Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct.    |

#### 13.2.2 PidStatus

PidStatus state of PID

**struct**

|               |                                                                           |
|---------------|---------------------------------------------------------------------------|
| serviceId     | <b>list</b> of <b>int</b><br>List of serviceIds a PID belong to.          |
| componentType | <b>optional component.ComponentType</b><br>ComponentType of PID           |
| ccErrors      | <b>int</b><br>CC errors on PID                                            |
| dataRate      | <b>bigint</b><br>Data rate of PID                                         |
| scrambling    | <b>Scrambling</b><br>Scrambling of PID (none, even or odd)                |
| PCR           | <b>bool</b>                                                               |
| present       | <b>bool</b><br>True if signalled pid has rate                             |
| language      | <b>optional string</b><br>Language of PID, if applicable (e.g. audio pid) |
| casId         | <b>optional int</b><br>Id of CAS system, if applicable (e.g. ECM pid)     |
| pidIn         | <b>int</b><br>PID number on input                                         |
|               | <b>optional int</b>                                                       |

|        |                      |
|--------|----------------------|
| pidOut | PID number on output |
|--------|----------------------|

### 13.2.3 Scrambling

Scrambling state of PID

#### enum

|      |                                   |
|------|-----------------------------------|
| NONE | No scrambling                     |
| EVEN | Scrambling with even control word |
| ODD  | Scrambling with odd control word  |

### 13.2.4 ServiceSourceStatus

Status for an input service

#### struct

|             |                                                       |
|-------------|-------------------------------------------------------|
| serviceName | <b>optional string</b><br>Service name                |
| pmtPid      | <b>int</b><br>PMT pid for service                     |
| pcrPid      | <b>optional int</b><br>PCR pid for service if present |

### 13.2.5 TransportStreamStatus

Status for the TS

#### struct

|                   |                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| ccErrors          | <b>int</b><br>Total CC error count for this TS.                                                                                                  |
| services          | <b>map</b> from <b>int</b> to <b>ServiceSourceStatus</b><br>Status for each service in the TS. Map from service ID to status struct.             |
| pids              | <b>map</b> from <b>int</b> to <b>PidStatus</b><br>Info on all PIDs, both signalled and unsignalled. Map from service PID value to status struct. |
| tsId              | <b>optional int</b><br>Transport Stream ID; set if PAT pid present in stream                                                                     |
| originalNetworkId | <b>optional int</b><br>Original Network ID; set if SDT pid present in stream                                                                     |